Quasi-Poisson models

Last time: modeling article publication

We are interested in analyzing the number of articles published by biochemistry PhD students. The data contains the following variables:

- `art`: articles published in last three years of Ph.D.
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program
- `ment`: articles published by their mentor in last three years

Activity

Work on the class activity (course website).

Recap: confidence interval

Poisson output:

```
                Estimate  Std. Error    z value     Pr(>|z|
(Intercept)  0.276493292 0.087294456  3.1673637 1.538278e-0
phd          0.008071036 0.026357684  0.3062119 7.594433e-0
ment         0.026537317 0.002000733 13.2637987 3.753884e-4
kid5        -0.099696463 0.035248105 -2.8284205 4.677832e-0
```

Want: 95% confidence interval for the change in the expected number of articles published associated with an increase of one in the number of children under age 6, holding PhD prestige and mentor publications fixed.

Activity: comparing standard errors

- Poisson estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{poisson}} = 0.03525$
- Bootstrap estimate: $\widehat{SE}(\widehat{\beta}_3)_{\text{boot}} = 0.0490$

> What will be the effect if we use the Poisson SE, when the Poisson distribution is not a good choice?

Overdispersion

Overdispersion occurs when the response Y has higher variance than we would expect if Y followed a Poisson distribution.

Let

$$\phi = \frac{\text{Variance}}{\text{Mean}}$$

> What should ϕ be if there is no overdispersion?

Estimating overdispersion

The *Pearson residual* for observation i is defined as

$$e_{(P)i} = \frac{Y_i - \widehat{\lambda}_i}{\sqrt{\widehat{\lambda}_i}}$$

$$\widehat{\phi} = \frac{\sum_{i=1}^{n} e_{(P)i}^2}{n - p}$$

- $p =$ number of parameters in model

Example: Estimating overdispersion

```
# fit the model
m1 <- glm(art ~ phd + ment + kid5,
          data = articles, family = poisson)

# get Pearson residuals
pearson_resids <- resid(m1, "pearson")

# estimate dispersion parameter
phihat <- sum(pearson_resids^2)/(915 - 6)
phihat
```

```
## [1] 1.888732
```

Handling overdispersion

Overdispersion is a problem because our standard errors (for confidence intervals and hypothesis tests) are too low.

> If we think there is overdispersion, what should we do?

Adjusting the standard error

- In our data, $\widehat{\phi} = 1.89$
- This means our variance is 1.89 times bigger than it should be
- So our standard error is $\sqrt{1.89} = 1.375$ times bigger than it should be

Adjusting the standard error in R

```
m2 <- glm(art ~ phd + ment + kid5, data = articles,
          family = quasipoisson)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *

(Dispersion parameter for quasipoisson family taken to be 1.88461)
```

- Allowing ϕ to be different from 1 means we are using a *quasi-likelihood* (in this case, a *quasi-Poisson*)

Adjusting the standard error

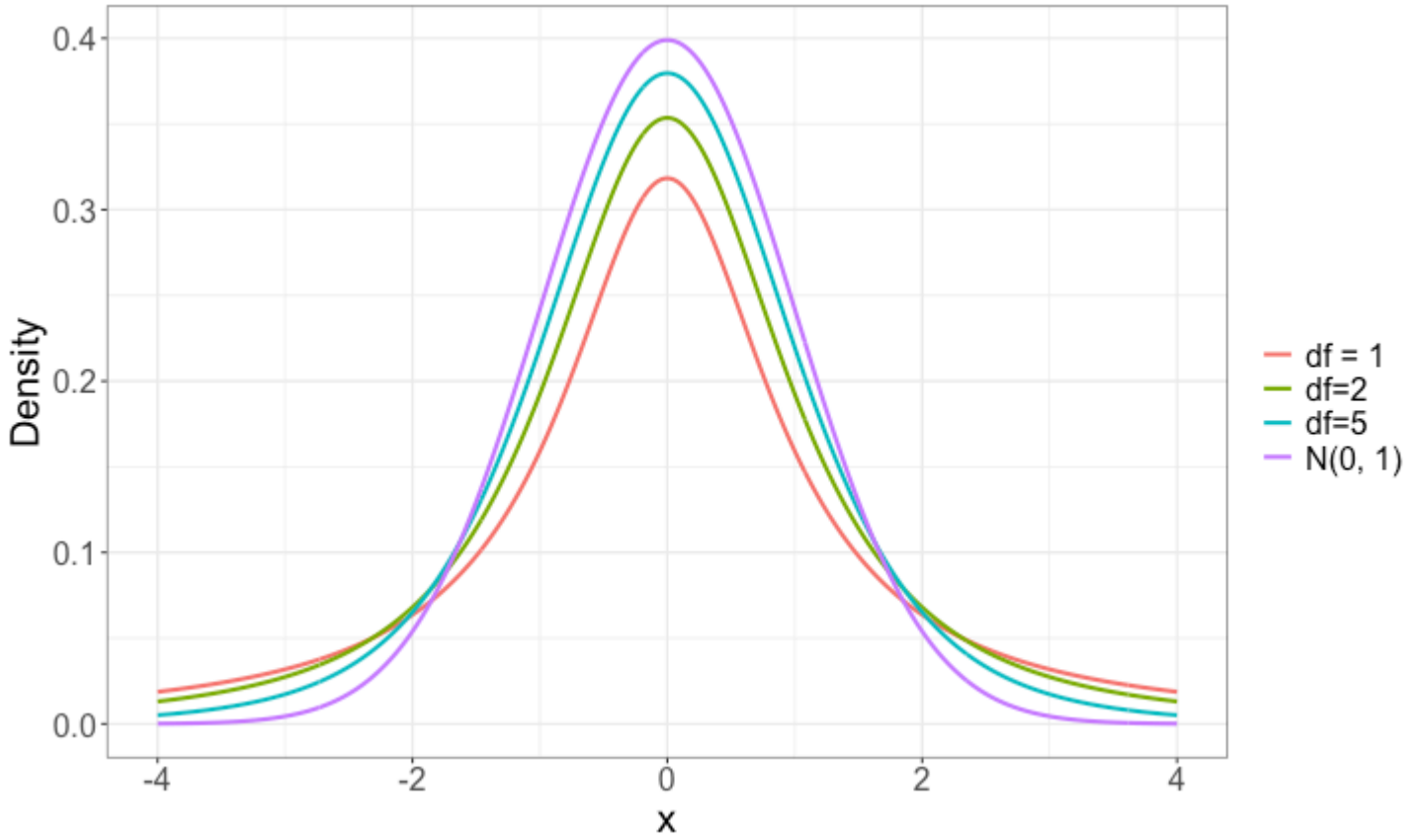
```
glm(art ~ phd + ment + kid5, family = poisson, data = articles)
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
(Dispersion parameter for poisson family
   taken to be 1)
```

```
glm(art ~ phd + ment + kid5, family = quasipoisson, data = articles)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
(Dispersion parameter for quasipoisson family
   taken to be 1.88461)
```

t-distribution



Calculating a confidence interval

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
```

New confidence interval for β_3:

$$-0.0997 \pm t^*_{n-p} \cdot 0.0484$$

```
qt(0.025, df=909, lower.tail=F)
```

```
## [1] 1.962577
```

$$-0.0997 \pm 1.963 \cdot 0.0484 = (-0.195,\ 0.005)$$